Hitesh Khatana

Full-Stack Engineer | GenAI Systems

hiteshkhatana1454@gmail.com – +91-96674-53353 – New Delhi, India

GitHub – LinkedIn

Professional Summary

Full-stack engineer with **4+ years** building production systems end-to-end, from **real-time backends at 20k–25k events/second** and customer-facing dashboards to **self-hosted Generative AI platforms**. Currently own an on-premise **RAG system at the Delhi High Court** that answers questions over judicial records with page-level citations. Owns products across the stack: API design, data pipelines, frontend, LLM integration, testing, and containerized deployment.

Technical Skills

Backend / APIs: Python, JavaScript, SQL, Django, FastAPI, Node.js, REST APIs, Microservices, Multi-Tenant SaaS, JWT Auth, RBAC, Kafka, Real-Time Event Processing.

Frontend: Next.js, React, TypeScript, JavaScript, Admin & Analytics Dashboards, Client-Side Tracking Pixels.

GenAI / LLM Systems: RAG, Prompt Engineering, Embeddings, Semantic & Vector Search, Chunking Strategy, Top-K Retrieval, Hallucination Mitigation, Citation-Grounded Generation, Hugging Face Transformers, PyTorch, sentence-transformers (MiniLM, mxbai), Ollama, Qwen3, Gemma 3, Llama 3.2, On-Premise / Offline LLM Deployment, RAG Evaluation (Recall@K, Groundedness), Question Regression Suites.

Data / Search / Infra: PostgreSQL, pgvector, ChromaDB, MySQL, Cassandra, SQLite, SQLAlchemy, Alembic, Redis, Elasticsearch, Kibana, Docker, Docker Compose, AWS (EC2, RDS, S3, SQS, CloudWatch), GitHub Actions (CI/CD), Git, Linux, pytest, pdfplumber.

Professional Experience

Full-Stack Engineer (Assistant Programmer) | Delhi High Court, New Delhi *Jan 2026 – Present*

Own an offline GenAI platform end-to-end on court hardware; data-sovereignty rules prohibit cloud AI APIs.

- Built and own an **offline RAG system** that answers questions across past judgments with the **exact page and paragraph cited**, running entirely on court hardware using **Node.js, React.js, ChromaDB, Ollama, and gemma3:12b**.
- Own the full pipeline end-to-end: **judgment PDF parsing, semantic chunking, embeddings, vector retrieval, prompting**, and the **REST APIs** that plug the system into existing court workflows.
- Improved retrieval relevance with **legal-structure-aware chunking** (facts, issues, reasoning, final order), so reasoning on one issue stays intact instead of being cut at a fixed length.
- Cut hallucinations by **enforcing grounding**: the system returns **NOT FOUND** unless the judgment text supports the answer, and every case name and page reference is verified against the source before display.
- Run a **question regression suite** across different legal query types, logging sources per query so any answer can be **audited later**.

Associate Product Software Engineer | SecureLayer7 / Sensfrx, Pune *May 2022 – Nov 2025*

Real-time fraud detection SaaS used by 250+ customer websites; worked across ML, DevOps, and QA teams.

- Built and maintained the backend of a **real-time fraud detection SaaS** serving **250+ customer websites** using **Django and Node.js**.
- Owned the **real-time scoring path**, handling **20k–25k user activity events per second** at **150–250 ms end-to-end latency**.
- Kept the pipeline fast at high volume using **Kafka and Redis** between services, **Elasticsearch and Kibana** for monitoring, and **MySQL and Cassandra** for activity storage.
- Built the **customer-facing dashboard** for fraud events and risk scores, and the **JavaScript tracking pixel** that collects user activity from client websites, owning the flow from browser to backend to UI.

Key Projects

- **EduLead AI | Multi-Tenant GenAI SaaS** (GitHub): Full-stack B2B platform where coaching institutes deploy an LLM chatbot grounded in their own documents. Built a **tenant-isolated RAG pipeline** on PostgreSQL + pgvector with all-MiniLM-L6-v2 embeddings and local Qwen3 inference via Hugging Face Transformers. Designed a **ContextBuilder** for prompt construction and **JWT/RBAC auth** enforcing tenant-scoped retrieval end-to-end. Shipped a **Next.js admin dashboard** for knowledge-base management and a **pytest suite** covering auth, tenant isolation, and RAG retrieval with a mocked AI service. Roadmap: RAG evaluation (Recall@K, groundedness), agentic lead-scoring workflows, AWS + Terraform deployment. *(Next.js, FastAPI, PostgreSQL, pgvector, Qwen3)*

Education & Awards

Maharshi Dayanand University, Rohtak — B.Tech in Computer Science & Engineering *2016 – 2020*

Hall of Fame Award (2024) and **Star Performer Award** (2023), SecureLayer7 / Sensfrx.